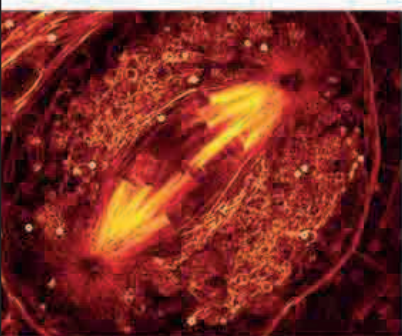


Close to the Earth's surface is the orange-red glow of the troposphere, which contains breathable air and our weather systems. The brownish layer, the tropopause, marks the transition to the gray-blue stratosphere beyond.



Micrograph showing cell division
As a cell divides, chromosomes are pulled apart, and the genes they carry are passed on to the two new cells formed.

ON JANUARY 1, NATHAN STUBBLEFIELD (1860–1928), a Kentucky farmer and inventor, showed off an electrical device that could send voice and music wirelessly over a distance of half a mile. Although it provoked scientific discussion, this wireless technology did not survive because it relied on disturbances generated by electromagnetic induction,

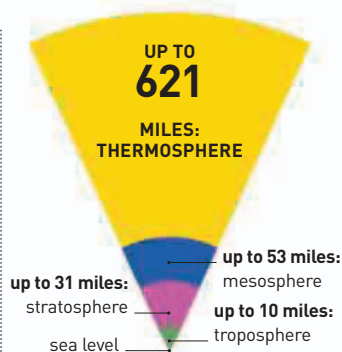
“INDIVIDUAL CHROMOSOMES POSSESS DIFFERENT QUALITIES.”

Theodor Boveri, *On Multipolar Mitosis as a Means of the Cell Analysis*, February 17, 1902

rather than radio signals, and was vulnerable to interference.

American and German biologists **Walter Sutton** (1877–1916) and **Theodor Boveri** (1862–1915) independently identified chromosomes as the carriers of genetic material.

Nearly 40 years earlier, Gregor Mendel had shown that inherited characteristics were the result of particles (see 1866). Sutton looked at sperm-forming cells of grasshoppers and saw that moving chromosomes mirrored Mendel's particles of inheritance. Boveri saw that sea urchin embryos needed an intact chromosome set to develop properly.



Layers of Earth's atmosphere
The atmosphere is made up of four layers. The gases are concentrated in the thin troposphere.

In April, meteorologist **Léon Teisserenc de Bort** (1855–1913) reported to the French Academy of Sciences on his investigation

of the atmosphere. Over a period of 10 years he had sent up more than 200 specially equipped hydrogen balloons. He found that weather systems occurred in a layer that extended at least 6 miles (9 km) above the Earth's surface. Beyond this layer the air was thinner and conditions calmer. Later, de Bort called the lower layer the troposphere, and the upper one, the stratosphere.

On February 17, the Stanley Motor Carriage Company was founded in the US for the production of a steam-driven car first built in 1897; the factory closed in 1924.

The Stanley Steamer
Early Stanley cars had boilers under the seats that generated steam. Known as Stanley Steamers, they were fuelled by a gasoline burner and started by a crank.

THE FIRST DESCRIPTION OF A PROCEDURE that would eventually become known as **chromatography** was presented at the Proceedings of the Warsaw Society of Naturalists in 1903. Russian botanist **Mikhail Tsvet** (1872–1919) had managed to separate the chemical components of plant pigments. He first let the mixture dissolve in petroleum ether and then ran it through a column of finely ground calcium carbonate. The orange, yellow, and green pigments separated into different bands: the ones that dissolved better in the solvent travelled faster and further. Tsvet's technique would later be adopted as an important analytical tool for separating mixtures of substances.

Humankind's attempts at powered flight reached an important breakthrough on **December 17**, when American inventor brothers **Orville** (1871–1948) and **Wilbur Wright** (1867–1912) achieved the first controlled man-carrying flight powered by an engine. For many years pioneers of aviation had tried hot-air balloons and gliders—with varying degrees of success. The



The Wright brothers' 1903 Flyer first flew in Kitty Hawk, North Carolina



January 1
Nathan Stubblefield demonstrates a new device that sends sound wirelessly

February 17
Theodor Boveri shows that an intact set of chromosomes is needed for normal development

American paleontologist **Barnum Brown** documented first remains of *Tyrannosaurus rex*

September
Ernest Rutherford and Frederick Soddy set out their theory that decay of atoms changes an element into another form

February
Ernest Rutherford calls Paul Villard's radiation gamma rays

June 21
Dutch physiologist **Willem Einthoven** describes first electrocardiograph

February 17
Stanley Motor Carriage Company is established

April 28
Léon Teisserenc de Bort describes stratosphere

Austrian biologists **Alfred von Decastello** and **Adriano Sturli** describe blood group AB

October 17
Walter Sutton suggests that chromosomes are the physical basis of Mendel's laws of inheritance

December
Robert Scott's Antarctic expedition arrives at the southernmost point reached by humans

March 21
Mikhail Tsvet introduces chromatography

Russian scientist **Konstantin Tsiolkovsky** describes how rockets could study outer space